

ProbFuse: A Domain-Decomposed Multi-Kernel Gaussian
Process for County-Level Agricultural Risk Assessment and
Conservation Policy Evaluation in the Maumee River Basin

A Thesis

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Abstract

This thesis presents ProbFuse, a probabilistic framework for agricultural risk assessment and conservation policy evaluation in the Ohio portion of the Maumee River Basin, a nutrient-impaired watershed feeding western Lake Erie. It addresses how to identify high-risk areas, areas where conservation investment has accumulated, and where yield-based policy tools are appropriate for future investment decisions.

The first contribution is a domain-decomposed multi-kernel Gaussian Process (GP) model for county-level corn and soybean yield anomaly prediction. Unlike a single isotropic kernel, the model uses separate kernels for climate, agronomic, and soil features. Under Leave-One-County-Out cross-validation across sixteen counties, the model achieves $\text{RMSE} = 12.56$ bu/ac, $\text{CRPS} = 6.95$ bu/ac, and $\text{Coverage@90} = 91.9\%$, with calibrated predictive intervals suitable for risk quantification. It also supports four-year prospective prediction for early-season outlooks.

The second contribution is a policy evaluation framework derived from the GP predictive distribution and historical NRCS funding records. The Agricultural Risk Index (ARI) combines predicted yield shortfall and uncertainty into a county-year vulnerability score. The Conservation Effectiveness Score (CES) captures lagged NRCS investment using acreage-normalized funding with decaying temporal weights.

The third contribution is a county agricultural system classifier based on USDA Census of Agriculture county profile data. By comparing commodity sales shares and livestock inventory indicators, the framework distinguishes livestock-dominant, row-crop-dominant, and mixed counties, ensuring the policy metrics are applied within scope.

Together, these contributions form a calibrated, scope-bounded framework for county-level agricultural risk assessment and conservation policy analysis.

To my family, with love and gratitude.

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Table of Contents

	Page
Abstract	ii
Dedication	iv
Acknowledgments	v
Vita	vii
List of Tables	xi
List of Figures	xiii
1. Introduction	1
1.1 Background	1
1.2 Problem Statement	3
1.3 Motivation	5
1.4 Research Questions	6
1.5 Thesis Contributions	7
1.6 Significance of the Study	9
1.7 Thesis Statement	10
1.8 Thesis Structure	11
2. Literature Review	12
2.1 Crop Yield Prediction and Uncertainty Quantification	12

2.2	Gaussian Processes for Agricultural Modeling	15
2.3	Conservation Practice Adoption and Program Evaluation	20
2.4	Remote Sensing and Conservation Monitoring	22
2.5	Soil Data and Spatial Interpolation	23
2.6	Research Gaps and Thesis Positioning	24
3.	Data	26
3.1	Study Area and Temporal Scope	26
3.2	Climate Data	28
3.3	Crop Yield Data	30
3.3.1	USDA Census of Agriculture Data	32
3.4	Remote Sensing: Off-Season NDVI	32
3.5	Soil Data	33
3.6	NRCS Conservation Practice Funding Data	35
3.7	Feature Engineering	36
3.8	Dataset Summary	37
4.	The Domain-Decomposed Multi-Kernel Gaussian Process	38
4.1	Overview	38
4.2	Kernel Selection by Negative Log Marginal Likelihood	40
4.3	Data-Driven Hyperparameter Initialization	42
4.4	Final Learned Kernel	44
4.5	Baseline Models and Experimental Setup	45
4.6	Model Evaluation Strategy	46
4.7	Summary	47
5.	Model Evaluation	48
5.1	Baseline Comparison	48
5.2	Leave-One-County-Out Evaluation	50
5.3	Calibration and Probabilistic Performance	51
5.4	Performance by Year Type	52
5.5	Temporal Horizon Evaluation	53
5.6	Summary	55

6.	Conservation Policy Analysis	56
6.1	From Model Outputs to Policy Metrics	56
6.2	Agricultural Risk Index	57
6.3	County Agricultural System Classification	58
6.4	Conservation Effectiveness Score	59
6.5	Policy Event Study	61
6.6	Conservation-Agriculture Gap Quadrant	63
6.7	Interpretation of County Classes	65
6.8	Summary	66
7.	Conclusion	68
7.1	Summary of Contributions	68
7.2	Limitations	70
7.3	Future Work	72
7.4	Conclusion	74

List of Tables

Table		Page
2.1	Representative modeling approaches and their relevance to this thesis.	19
3.1	Summary of data sources, temporal coverage, and derived features. .	37
4.1	Sub-kernel structure selection results. NLML is the primary model-selection criterion; RMSE and CRPS are reported on the held-out test set for reference. Lower NLML, RMSE, and CRPS are better. Bold indicates the selected kernel for each group.	42
4.2	Hyperparameter initialization and sensitivity analysis. The final solution is robust to moderate perturbations of the noise initialization and optimizer restart count.	44
5.1	Baseline comparison under temporal split (test: 2018-2022, $n = 111$ observations). Lower RMSE and CRPS are better; higher R^2 and Coverage@90 are better. R^2 is not reported for GP models because the primary evaluation criteria are probabilistic (CRPS and Coverage@90)	48
5.2	Leave-One-County-Out evaluation results for the multi-kernel GP. Coverage@90 is the empirical fraction of test observations falling inside the model's 90% prediction interval. The final row reports the basin-level mean.	50
5.3	LOCO performance stratified by year type. Dry years: SPEI-12 < -0.5 ; wet years: SPEI-12 $> +0.5$; normal years otherwise.	53

5.4	Prospective temporal horizon evaluation. Each row represents a forecast horizon h years, with training restricted to years up to $2022 - h$ and testing on the subsequent h years.	54
6.1	County agricultural system classification based on USDA NASS 2022 County Profile reports. Classification reflects share of sales by type and livestock inventory indicators from the official county profiles. . .	60

List of Figures

Figure	Page
3.1 Study area map of the Ohio portion of the Maumee River Basin, showing county boundaries and the counties retained for the primary analysis.	27
3.2 Temporal coverage of the five data sources used in the thesis, showing the overlap between climate, yield, NDVI, soil, and NRCS funding records.	31
3.3 County-level yield anomaly trends for corn across the Maumee Basin study period.	35
4.1 Conceptual structure of the domain-decomposed multi-kernel Gaussian Process. Separate kernels are assigned to climate, agronomic, and soil/spatial feature groups. The resulting covariance components are added to form the final GP prior. The design reflects the fact that each feature domain has a different smoothness structure and a different agronomic interpretation.	40
6.1 Lag weights used in the Conservation Effectiveness Score. The scheme assigns the greatest credit to the most recent year and progressively discounts older investments, reflecting the delayed accumulation of soil health and conservation practice benefits.	62
6.2 Conservation-Agriculture Gap quadrant classification. Counties are positioned by their normalized ARI and CES values relative to basin medians, distinguishing priority counties, supported-but-struggling counties, resilient counties, and lower-need counties.	65

Chapter 1: Introduction

1.1 Background

The Maumee River Basin is one of the most heavily managed agricultural watersheds in the United States and is in a critical location in the context of both Great Lakes resource conservation and nutrient management policy. The basin covers parts of Ohio, Indiana, and Michigan and flows into the western basin of Lake Erie, where high levels of phosphorus have for several decades been linked to damaging algal blooms, hypoxic dead zones, negative impacts on drinking water quality, and escalating economic costs to local communities [28, 29, 12]. Within the basin, row-crop agriculture (predominantly corn and soybeans) dominates land use. The interaction between crop production systems, subsurface tile drainage, soil characteristics, and year-to-year climate variability makes agricultural outcomes highly spatially and temporally heterogeneous [12, 28].

Whereas the debate over water quality in Lake Erie is dominated by the topics of nutrient load and algal bloom levels, the more useful framework for the present thesis

is the implementation of conservation practices and the effectiveness of conservation practices in achieving positive outcomes. This is the type of question that existing methods do not answer well.

Conservation programs operating within the Maumee Basin are largely implemented through the NRCS working-lands programs including Environmental Quality Incentives Program (EQIP) and Conservation Stewardship Program (CSP)[16]. Working lands programs are different from land retirement programs in that they require conservation practices to be implemented within active agricultural production rather than being implemented to retire land from production. In practice, however, implementation of the programs is limited by application pressure, staff limitations, and the difficulty of linking practices to the individualized risk profile of each county [6, 7, 8].

A further complication is that conservation benefits are typically delayed. Practices such as cover crops, nutrient management plans, and reduced or no-till cultivation may affect yield stability and soil health over multiple seasons rather than producing an immediate response in the year of adoption [15, 14]. This lag structure creates a fundamental evaluation problem. If program analysts treat annual funding as a one-year treatment event, they risk missing the cumulative effect of sustained investment and may incorrectly classify counties that are early in their conservation

trajectory as program failures. The reverse error is also possible: counties with long-standing support may appear underwhelming if the evaluation window is too short or if the analysis does not account for the time required for benefits to emerge.

In this context, NRCS district conservationists and Technical Service Providers need an analytical signal to determine where conservation investment is needed, where benefits are accumulating, and where resiliency has been achieved. The Maumee Basin is an interesting location in which to perform such work. In fact, it has an unusually long and consistent record of publicly available data.

1.2 Problem Statement

The main challenge addressed by this thesis is the lack of a county-level, uncertainty-aware tool able to simultaneously quantify agricultural yield risk, summarize conservation investment, and combine these two measures into a useful signal directly applicable to NRCS operations.

Current conservation program assessment techniques are typically limited to summaries of total dollars obligated, total contracts signed, or total acres treated, but do not address the critical question of whether investment is matched to county-level risk or whether investment is effective, or both, over time. This is significant for at least three reasons:

First, county-level crop yield models are typically either deterministic or only weakly stochastic. While a predicted value may be useful, it does not convey information on how confident the model is in its predicted value, and this information

is critical when making conservation investment decisions. For example, a county-level forecast might indicate a county is expected to fall 10 bushels per acre below its long-run trend, but this is a very different policy question if this forecast has a narrow or broad uncertainty.

Second, conservation benefits accumulate gradually and unevenly across time. A framework that treats conservation as a one-year binary event misses the cumulative dynamics of soil health investment and is likely to either underestimate the benefit of sustained support or overestimate the effect of recent and still-maturing practices. A useful evaluation framework must be able to distinguish between initial adoption, active accumulation, and matured benefit.

Third, no standard county-level metric currently integrates agricultural risk with conservation investment history in a way that is directly actionable for field-level program management. In the absence of such a metric, counties that are both highly vulnerable and historically under-supported cannot be systematically distinguished from counties that are similarly vulnerable but have already received sustained investment that may be approaching its period of maximum effect.

This thesis addresses all three gaps through a single integrated framework. It does not seek only to improve yield prediction accuracy. Rather, it uses the probabilistic GP model as the foundation for a policy analysis layer whose outputs are designed to be interpretable and useful for NRCS practitioners.

1.3 Motivation

The primary motivation for this thesis is to support a more principled evaluation of conservation funding in agricultural watersheds. NRCS district conservationists and related practitioners are frequently asked to make program decisions under uncertainty: which counties should receive prioritized outreach, where technical assistance should be concentrated, and whether program resources appear to be reaching the places where conservation need is greatest. Annual funding totals are not sufficient information on which to make investment decisions because they do not convey information on whether the county being funded is currently at risk, whether conservation practices are mature enough to begin demonstrating benefits, or whether recent investment has had sufficient time to do so.

The Maumee Basin is a particularly well-suited location for this type of investigation because it sits at the intersection of agricultural production and environmental concern. It has been repeatedly targeted by state and federal agencies for nutrient reduction and agricultural conservation efforts [29, 28]. It has also been the focus of substantial policy attention related to phosphorus export and Lake Erie water quality. At the same time, it remains a major agricultural production region where conservation must operate within an existing working landscape. These conditions make it especially well suited for evaluating conservation effectiveness, rather than merely documenting the presence of conservation investment.

Another impetus for this thesis is the availability of quality long-term data sets. There are county-level agricultural data sets related to crop yields that go back to the 1970s. There are long-term climate records available from NOAA weather stations. There are soil data sets from SSURGO. There are off-season satellite images that can be used as a proxy for continuous data related to vegetative cover. Such data sets would be useful for creating a model that is not only statistically sound but also accessible. Such a model would be particularly useful for a machine learning thesis. Machine learning models are not simply attempts to forecast data but to forecast data that can be connected back to some level of physical reality.

1.4 Research Questions

This thesis is organized around three research questions.

RQ1. (Technical accuracy and calibration). Can a domain-decomposed multi-kernel Gaussian Process, with sub-kernel structures selected by Negative Log Marginal Likelihood and hyperparameters initialized using data-driven heuristics, produce county-level crop yield predictions with calibrated uncertainty that outperform single-kernel probabilistic baselines under Leave-One-County-Out (LOCO) cross-validation and yield lower or comparable temporal-split RMSE than deterministic baselines.

RQ2. (Agricultural interpretation). Do the probabilistic model outputs capture a coherent spatiotemporal signal of agricultural risk (including differential performance under dry, normal, and wet conditions) that aligns with known patterns of climate stress and soil variability across the Maumee Basin counties?

RQ3. (Policy effectiveness). Does lagged NRCS conservation investment, summarized through the Conservation Effectiveness Score, predict measurable changes in county-level agricultural risk over time in a way that is consistent with the documented delayed onset of soil health and cover crop benefits?

1.5 Thesis Contributions

This thesis makes three interconnected contributions to machine learning, agricultural modeling, and conservation policy evaluation.

Technical & Methodological contribution. A domain-decomposed multi-kernel Gaussian Process model for county-level corn and soybean yield-anomaly prediction, with sub-kernel structure selection via Negative Log Marginal Likelihood and hyperparameter initialization using data-driven heuristics. The feature domain is explicitly partitioned into climate, agronomic, and soil groups, each with its own covariance function, reflecting the structurally distinct physical behavior of each domain and preserving interpretability through domain-separated uncertainty decomposition. The model is evaluated under Leave-One-County-Out cross-validation

and prospective temporal holdout, with emphasis on calibrated probabilistic performance.

Agricultural contribution. A long-run probabilistic approach for understanding county-level yield risk in the Maumee Basin from 1986 to 2022, using a feature set including SPEI-based drought indices, growing degree days, off-season NDVI as a proxy for cover intensity, lagged agronomic states, and SSURGO-based soil attributes. This provides a means of understanding agricultural vulnerability at the county level over long periods rather than only year-to-year anomalies.

Policy contribution. A county-level conservation evaluation system linking historical investment records from NRCS with model-derived risk estimates, including the Agricultural Risk Index, Conservation Effectiveness Score, and Conservation-Agriculture Gap classification. These indices are intended to help NRCS practitioners identify where investment appears insufficient, where it may be accumulating toward delayed benefit, and where additional outreach is likely to be most valuable.

The primary technical novelty of this thesis is the domain-decomposed kernel design itself: assigning structurally distinct covariance functions to climate, agronomic, and soil feature groups, selected by NLML, is a principled design choice not previously applied in county-level agricultural GP modeling. The primary application novelty is the end-to-end integration of that probabilistic model output with NRCS investment history into actionable policy indices designed for use by NRCS

practitioners. The two are distinct contributions: the technical design is what makes the predictive distributions reliable; the application framework is what makes them useful.

1.6 Significance of the Study

The immediate audience for this work is NRCS district conservationists and Technical Service Providers, who need a county-scale view of conservation effectiveness beyond aggregate funding totals. A framework that combines calibrated yield prediction with conservation investment history can support more informed decisions about outreach priorities, program targeting, and long-term resilience planning. It can also help practitioners distinguish between counties that are genuinely underserved and counties that are already receiving sustained support but may need time for benefits to fully emerge (a distinction that is currently difficult to make with existing data and tools).

Beyond NRCS program staff, several other groups stand to benefit from this framework. Farmers and producers in the Maumee Basin gain a county-level resilience signal that helps contextualize when soil-health investments like cover crops, no-till, and nutrient management are likely to translate into stabilized yields, given the empirically observed four-year benefit onset lag. Commodity-focused crop analysts and supply-chain practitioners gain access to calibrated probabilistic yield anomaly forecasts that are more useful for risk assessment than deterministic point

predictions precisely because they convey both the expected outcome and the confidence in that expectation. Finally, the framework is designed as a reproducible template: researchers and educators working on other nutrient-impaired watersheds can adapt the domain-decomposed GP and the ARI-CES pipeline to their own regional datasets without starting from scratch.

From a broader computational perspective, this thesis demonstrates how uncertainty-aware machine learning can be integrated into a policy evaluation setting without collapsing the problem into a purely abstract forecasting exercise. The models here are not intended to replace field expertise. Rather, they provide an additional quantitative signal that practitioners can incorporate into existing decision workflows. Because the framework is county-level and built from publicly accessible data sources, it can be adapted to other agricultural watersheds facing similar questions about risk targeting and conservation investment effectiveness.

1.7 Thesis Statement

This thesis argues that a domain-decomposed multi-kernel Gaussian Process, trained on long-run county-level climate, agronomic, soil, and remote-sensing data from the Maumee River Basin, can produce calibrated probabilistic predictions of crop yield that are sufficiently reliable and interpretable to serve as the technical foundation for county-level policy indices that evaluate conservation investment effectiveness and support more principled NRCS conservation targeting in nutrient-impaired agricultural watersheds.

1.8 Thesis Structure

The remainder of this thesis is organized as follows. Chapter 2 reviews the relevant literature on crop yield prediction, Gaussian Process modeling, conservation practice adoption, remote sensing, and soil data. Chapter 3 describes the data sources and preprocessing. Chapter 4 presents the Gaussian Process methodology. Chapter 5 evaluates the model. Chapter 6 develops the conservation policy indices. Chapter 7 concludes the thesis.

Chapter 2: Literature Review

This thesis draws on three research streams that have often developed separately: crop yield prediction and uncertainty quantification, Gaussian Process modeling for agricultural and environmental applications, and conservation program evaluation in Midwestern agricultural systems. This chapter reviews each of these streams, identifies their limitations, and places the work presented within this thesis within context.

2.1 Crop Yield Prediction and Uncertainty Quantification

Crop yield prediction has been an active area of research for decades, with models ranging from classical statistical regression techniques through to more recent developments in deep learning architectures. Early work focused on ordinary least squares and related linear models, seeking to quantify the relationship between annual yields and seasonal climate aggregates, e.g., growing season precipitation and degree days [30]. These models are attractive for their interpretability and low data demands,

but established an important empirical fact: interannual variability in weather explains a significant fraction of county-level yield variation within the Midwest. While highly effective, their primary limitation for policy-relevant work is their inability, within a linear model, to quantify nonlinear relationships between drought stress, soil, and management practices, as are observed within actual agricultural systems.

Machine learning methods have improved point-prediction accuracy by relaxing linearity assumptions between predictors and yield. Random Forest, Gradient Boosting, and Support Vector Machine algorithms have been commonly used for predicting yield and have been successful in achieving higher accuracy than other methods in terms of mean squared error for point prediction when climate, remote sensing, and soil data are used together [11, 9]. The major drawback of these methods is that they are point predictors and cannot be used for providing prediction intervals.

The practical importance of this limitation is easy to understate but important to note. Conservation targeting is not a problem of minimizing expected loss over many decisions; it is a problem of making a single decision for each county in the face of uncertainty. A county that is forecast to be 10 bushels per acre below trend level has different implications depending on whether the uncertainty interval is small or large. Without accounting for uncertainty, a policy tool may appear precise while being fundamentally inaccurate for the decisions it is intended to guide.

Deep learning models have also been applied for yield prediction, especially for scenarios with access to remote sensing time series data. Convolutional Neural Networks and Long Short-Term Memory networks have shown strong potential for handling complex nonlinear dynamics and temporal dependencies. These models have reported strong results for yield estimation at the pixel level or field level [34, 32]. Even though they have shown strong results for yield estimation, deep learning models are generally less suitable for the current problem for several reasons: they require large amounts of labeled data for training and are generally not well suited for handling uncertainty estimates [5].

Process-based crop models such as DSSAT [10], APSIM, and EPIC have the advantage of providing a mechanistic alternative to statistical approaches. In process-based crop models, the bio-physical processes of crop development, water movement in the soil, and nutrient cycling are simulated based on physical laws. In process-based crop models, the relationships between conditions and outcomes are explicit and easily interpretable. However, process-based crop models also have the disadvantage of being highly data-intensive and requiring field-scale parameterization of planting dates, variety characteristics, irrigation practices, and nutrient applications, which is not publicly available for the county scale. In addition, the output of process-based crop models is usually deterministic unless the Monte Carlo method is applied. However, the complexity of the implementation of the Monte Carlo method is substantial. In the context of the present thesis, process-based crop models have

the advantage of providing a useful conceptual framework for the physical structure of crop yields. However, they have the disadvantage of not being a viable option for the present thesis.

The next set of approaches deals with uncertainty quantification. Bayesian approaches have the advantage of providing uncertainty quantification in the form of posterior distributions for the model parameters. However, the application of Bayesian approaches to crop models is computationally intensive [3]. Conformal prediction methods have the advantage of providing uncertainty quantification in the form of prediction intervals for any point predictor. However, the prediction intervals generated by the conformal prediction method do not depend on the internal representation of the model and might not be informative in regions of high uncertainty. Gaussian Processes have the advantage of being inherently probabilistic. The posterior predictive distribution is the output of the model rather than the input to the model.

The literature therefore reveals a gap: accurate point-prediction models exist, but county-scale calibrated uncertainty suitable for conservation policy use remains limited.

2.2 Gaussian Processes for Agricultural Modeling

Gaussian Processes (GPs) form a natural basis for probabilistic regression over tabular data, as they specify a prior distribution over functions and deliver a posterior predictive distribution at every new location [20]. In the GP formulation, the

prediction for a new county-year location is accompanied by a posterior mean (the best point prediction of the yield anomaly) and a posterior variance (a principled measure of the uncertainty remaining after conditioning on the available data). This is exactly the kind of output required by the policy framework in the present thesis.

A key advantage of GPs in agriculture is the interpretability of the covariance function. The kernel function encodes the structural assumptions about the mapping from the input space to the output space. The various input features operate on different scales and have different degrees of smoothness. Climate features vary abruptly from year to year, while soil features vary smoothly across geography and remain stable over time. In contrast, management features might have “carryover” effects with delayed response. A kernel-based model can capture each of these structures individually, rather than trying to put everything through a single covariance function.

The definitive reference for GP regression establishes that Negative Log Marginal Likelihood (NLML) is an appropriate criterion for joint model structure selection and hyperparameter estimation [20]. NLML balances the goodness of fit to the data and the complexity of the model through the log determinant term, thereby implementing Bayesian Occam’s Razor. This is particularly useful in the presence of limited training data, in which overfitting to noise features is a real concern.

Prior research has applied GPs in agricultural and environmental modeling in a variety of ways. Some studies use single-kernel GPs for spatial interpolation of soil or

climate variables, taking advantage of the kriging interpretation of GP regression [21]. Others embed GP components within hybrid architectures that combine deep feature extraction with a GP output layer [34]. These studies collectively show that the probabilistic structure of a GP is useful whenever uncertainty quantification matters. However, most existing agricultural GP applications use a single isotropic kernel that treats all input dimensions identically. That design choice is often too restrictive for agricultural systems because it ignores the distinct physical interpretation of different feature groups.

The multi-kernel approach in this thesis extends the additive kernel framework developed in the GP literature [4]. By partitioning features into domain-specific groups and assigning each group its own covariance function, the model can reflect the fact that climate stress, management carry-over effects, and soil buffering capacity are not the same type of signal and should not be modeled as if they were. Additive kernel structures also preserve interpretability: because covariance components add rather than interact multiplicatively, it is possible to reason about the relative contribution of each domain to the overall prediction.

The choice of specific kernel family within each domain is also theoretically motivated. Smooth kernels such as the Radial Basis Function (RBF) are appropriate for variables that are expected to vary continuously and slowly over the input space, such as soil properties and geographic coordinates [21]. Matérn kernels with finite smoothness parameters are more appropriate for signals that are structured but not

infinitely smooth; a description that fits climate indicators such as SPEI, which reflect weather patterns that are correlated across years but can change abruptly under extreme events [20]. The Matérn-3/2 and Matérn-5/2 kernels are once and twice differentiable, respectively, which means they can represent functions with localized discontinuities or near-discontinuities that a perfectly smooth RBF kernel would inevitably oversmooth.

Hyperparameter initialization is another important practical issue in GP modeling. The NLML objective is non-convex with respect to the hyperparameters, which means gradient-based optimization can converge to suboptimal local solutions if it is started too far from the true optimum. [2] show empirically that data-driven initialization strategies outperform arbitrary defaults across a range of kernel types. The median pairwise Euclidean distance in the scaled feature space is a widely recommended starting point for length scales because it places the model in a regime where a substantial fraction of training pairs have non-trivial covariance - neither nearly zero nor nearly one [2, 17]. The present thesis uses this heuristic for all three feature groups, together with variance-based initialization for signal and noise terms.

Finally, it is worth noting what is still missing in the GP agricultural modeling literature. While domain-decomposed or additive kernel designs have been studied theoretically [4], few published applications use them in county-level agricultural

prediction with explicit kernel structure selection and downstream policy interpretation. The present thesis fills that gap by combining principled kernel design with a county-level conservation policy framework.

Table 2.1 summarizes the representative modeling approaches reviewed in this section.

Table 2.1: Representative modeling approaches and their relevance to this thesis.

Approach	Strengths	Limitation for this thesis
Linear regression	Interpretable, simple, low data requirement	Cannot capture nonlinear interactions or produce calibrated uncertainty
Random Forest / Gradient boosting	Strong point prediction, handles nonlinearities	Limited probabilistic structure; no calibrated predictive intervals
Deep learning	Flexible, can exploit remote sensing time series	Requires large data; uncertainty estimates often poorly calibrated
Process-based models	Mechanistic and physically interpretable	Requires field-level inputs not available at the county scale
Single-kernel GP	Probabilistic, uncertainty-aware, principled	Isotropic kernel conflates climate, agronomic, and soil signal structures
Domain-decomposed GP (<i>this thesis</i>)	Calibrated uncertainty, domain-structured, interpretable	Computationally heavier; requires principled kernel design

2.3 Conservation Practice Adoption and Program Evaluation

The second research stream relevant to this thesis concerns conservation practice adoption and the evaluation of agricultural conservation programs in Midwestern working-lands contexts. Conservation practices such as cover crops, nutrient management plans, no-till, and reduced tillage are central to the policy toolkit for addressing both agricultural resilience and phosphorus export in the Maumee Basin [15, 14, 1]. Although the environmental rationale for these practices is well established, their implementation, persistence, and measurable effectiveness are more difficult to evaluate at the county scale.

Field-scale and watershed-scale studies provide the clearest evidence that conservation practices can substantially reduce erosion and nutrient losses when implemented consistently. USDA CEAP analyses of the western Lake Erie basin showed meaningful reductions in soil loss and edge-of-field nutrient transport under conservation scenarios compared to pre-program baselines [27]. More recent watershed studies have shown that cover crops, subsurface drainage management, and nutrient placement can reduce DRP losses even in anomalously wet years [24, 1].

However, adoption has not been uniform across the Midwest or within the Maumee Basin. Satellite-based research indicates that cover crop adoption has increased substantially since 2011, although it remains a minority of corn and soybean acres[22].

Adoption also varies substantially across counties in ways that are influenced by information access, producer characteristics, program outreach, and economic factors rather than biophysical need [19, 33]. A related problem is disadoption: conservation practices are not necessarily permanent and may be abandoned when costs, logistics, or changing farm conditions reduce their value [18]. This matters for program evaluation because historical funding may overstate current conservation benefit if practices are no longer maintained.

The literature also reveals structural limitations in program access. Conservation policy organization reports have found that demand for the program exceeds funding, approval of applications varies based on staff activity rather than producer need, and counties with higher baseline conservation capacity may be better able to compete for funding [6, 7, 8]. These studies suggest that county funding levels cannot be taken as a simple indicator of current need or program effectiveness, but also reflect access and capacity factors.

From the perspective of policy evaluation, this literature reveals the methodological need for a model capable of differentiating funding events from conservation capital, accommodating the lagged receipt of program benefits, and relating cumulative program investment to county yield outcomes. That is precisely the role of the CES metric developed in Chapter 6 of this thesis.

2.4 Remote Sensing and Conservation Monitoring

Remote sensing has become an important tool for monitoring agricultural management at regional scales because it provides spatially consistent, repeatable observations that do not depend on producer self-reporting. Vegetation indices, particularly the Normalized Difference Vegetation Index (NDVI), are especially useful for detecting off-season vegetative cover because cover crops produce a distinct greenness signal between harvest and spring planting that is largely absent from bare or poorly managed fields [22].

Recent work has shown that multi-source satellite data can be used to track cover crop adoption trends across large regions. Sun et al. [22] used a Landsat-MODIS fusion approach to estimate cover crop presence across the Midwest. This is directly relevant to this thesis because off-season NDVI is used as a continuous feature representing cover intensity, allowing the Gaussian Process model to learn how the level of vegetative cover relates to yield anomalies and downstream conservation performance.

The advantage of remote sensing in this context is that it captures management intensity as a gradient rather than a binary indicator. A continuous NDVI signal is better suited to capturing this variation than administrative enrollment records, which record whether a contract was signed but not how consistently the practice was implemented.

A practical limitation of the remote sensing approach used here is that county-level NDVI aggregates smooth over within-county heterogeneity. This is an inherent constraint of the county-level modeling framework and is discussed further as a direction for future integration with precision agriculture data.

2.5 Soil Data and Spatial Interpolation

Soil properties are a critical and often underweighted component of agricultural yield prediction at the county scale. Available water capacity, organic matter content, drainage class, and soil vulnerability collectively determine how a given county’s crop production responds to climate variability. In the Maumee Basin, where soils range from well-drained loams to poorly drained heavy clays overlying extensive tile drainage systems, soil heterogeneity is a first-order driver of both yield levels and yield risk.

The SSURGO database [13] provides a standardized, nationally consistent source of soil information at the map unit level. County-level attributes derived from SSURGO are widely used in agricultural modeling because they are consistently available across space and time. AWC is especially important in the present context because it mediates yield losses in dry years: high-AWC soils can buffer a crop through a short-term drought while low-AWC soils show rapid stress response. On the other hand, counties without sufficient natural drainage or tile infrastructure will have higher probability of waterlogging in wet years, thus depressing county yield below trend in anomalously wet years.

The conceptual link between soil properties and GP kernel design is the spatial correlation structure that characterizes soil variation. Classical geostatistics (kriging in particular) is built around the assumption that soil properties are spatially correlated, with nearby locations being more similar than distant ones [21]. This same assumption motivates the use of smooth covariance functions, such as the RBF kernel, for the soil feature group in this thesis. Soil properties change gradually across the landscape; they do not exhibit the abrupt year-to-year shifts that characterize climate anomalies. A smooth kernel reflects that structure, and the kernel selection procedure in Chapter 4 confirms that the RBF kernel achieves a better marginal likelihood than Matérn variants for the soil feature group.

2.6 Research Gaps and Thesis Positioning

The literature reviewed in this chapter points to four specific gaps that motivate the present work.

First, crop yield prediction literature contains many accurate point-prediction models but relatively few that produce calibrated uncertainty in a form suitable for conservation policy use. This is especially true at the county level, where sample sizes are modest and interpretability is important. The multi-kernel GP in this thesis is designed to address that gap by providing principled predictive distributions rather than point estimates.

Second, the Gaussian Process literature offers strong theoretical support for probabilistic agricultural prediction, but few published county-level applications use

domain-decomposed kernels to separate climate, agronomic, and soil structure in a principled way. The kernel design in this thesis is explicitly motivated by that gap, and the NLML-based kernel selection procedure provides an objective and reproducible justification for the structural choices made.

Third, the conservation program evaluation literature generally focuses on adoption, funding, or environmental outcome, but does not often relate county investment history to long-run yield risk in a lag structure. The CES metric developed in this thesis address that gap by encoding the temporal decay structure of conservation benefits directly into the policy metrics.

Fourth, many county-level agricultural analyses implicitly assume that all counties in a region are methodologically comparable, even when the agricultural system differs substantially. A yield-based risk framework is not appropriate for counties where livestock production dominates because yield anomaly is not a meaningful summary of agricultural output in those settings. The county agricultural system classifier developed in Chapter 6 addresses that gap by ensuring that the framework is applied only within a defensible and documented scope.

Together, these four gaps define a clear role for the integrated framework developed in this thesis: a county-level, probabilistic, lag-aware, scope-bounded framework for agricultural risk assessment and conservation policy evaluation in the Maumee River Basin.

Chapter 3: Data

This chapter describes the five core data sources used in this thesis, the preprocessing and quality-control pipeline applied to each, and the feature engineering decisions that produce the county-year feature matrix consumed by the Gaussian Process model. All data processing is implemented in Python and is fully reproducible from publicly available source files and web-accessible interfaces [16, 13, 31, 23].

3.1 Study Area and Temporal Scope

The study area is the Ohio portion of the Maumee River Basin, which comprises counties whose drainage area is predominantly contained within the Maumee watershed and its major tributaries. The Maumee watershed is a major agricultural basin spanning approximately 6,350 square miles in northwest Ohio, northeast Indiana, and southeast Michigan [28, 12]. The Ohio portion covers sixteen counties that span a gradient from more diversified agricultural systems in the eastern portion to highly specialized corn-soybean production in the western and central portions.

After applying the agricultural system classification filter described in Section 6.3, a row-crop-dominant subset of these counties is used for the primary policy analysis.

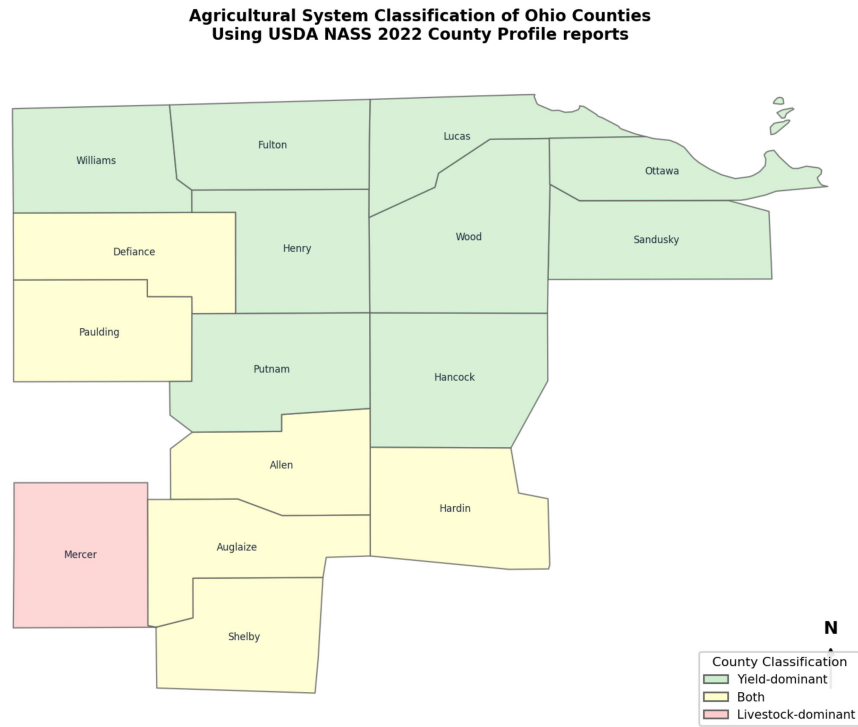


Figure 3.1: Study area map of the Ohio portion of the Maumee River Basin, showing county boundaries and the counties retained for the primary analysis.

The yield prediction model covers 1986–2022, the period for which county-level yield, climate, remote sensing, and soil data are simultaneously available. The model does not extend prior to 1986 because county-level NASS yield data, NOAA climate station data, and satellite image data will not be simultaneously available for the full

period for the counties in the Maumee basin. The policy analysis uses the shorter period 2014-2022 based on the availability of county-level NRCS practice funding data.

3.2 Climate Data

Daily climate observations are provided by the NOAA National Centers for Environmental Information (NCEI) land-based surface archive. Station-level records are assembled from two overlapping historical datasets and merged by station identity and observation date. Station coordinates are geocoded to assign each station to the county within whose boundary it falls, and county-level daily values are computed as the unweighted mean across all stations assigned to each county.

For each station-day record, precipitation and maximum and minimum temperature are extracted. These daily observations are aggregated to monthly totals for precipitation and monthly means for temperature extremes. Monthly potential evapotranspiration (PET) is estimated using the Thornthwaite method [23], which is a temperature-based approximation that is appropriate for data-sparse regional analysis where radiation and humidity records are incomplete. Monthly water balance is computed as precipitation minus PET, and the Standardized Precipitation-Evapotranspiration Index (SPEI) is derived at 3-, 6-, and 12-month accumulation windows [31].

SPEI is a multiscalar drought index that accounts for both the moisture supply (precipitation) and demand (evapotranspiration) dimensions of water balance, making it sensitive to temperature-driven drought as well as precipitation deficits. The three accumulation windows capture distinct aspects of agricultural water stress: SPEI-3 reflects short-term in-season moisture conditions relevant to immediate crop stress; SPEI-6 captures the seasonal cumulative water balance relevant to soil moisture recharge and growing season productivity; and SPEI-12 captures longer-run hydrological conditions relevant to multi-year carry-over effects and annual drought classification.

Additional climate features are computed from the daily temperature record. Growing degree days (GDD, base 10°C) are accumulated from April through September to capture the thermal environment available to corn and soybean development. Extreme heat days (EHD) are counted as the number of days within the growing season with maximum temperature exceeding 35°C, capturing heat stress that can reduce pollination success and grain fill. Growing-season precipitation (gs_precip) is the total precipitation from May through August. These six variables (SPEI-3, SPEI-6, SPEI-12, GDD, EHD, and gs_precip) constitute the climate feature group for the Gaussian Process model.

A county-year drought classification is also derived from SPEI-12 for use in year-type stratification in the evaluation results presented in Chapter 5: dry years are

defined as $\text{SPEI-12} < -0.5$, wet years as $\text{SPEI-12} > +0.5$, and normal years otherwise.

3.3 Crop Yield Data

County-level corn and soybean yield, planted acreage, and harvested acreage are sourced from the USDA National Agricultural Statistics Service (NASS) Quick Stats database [26]. These data are annual survey estimates reported at county scale, making them appropriate for a basin-wide longitudinal analysis. The NASS county program has maintained consistent survey methodology across the study period, supporting long-run comparisons.

The target variable is *yield anomaly*, defined as the deviation of the county’s reported yield from its county-specific long-run linear trend. The linear trend is estimated by ordinary least squares regression of yield on calendar year for each county-crop combination. Detrending removes the component of yield variation attributable to broad technological improvement over time (variety genetics, precision application, hybrid development) and isolates the year-to-year variability associated with weather stress, soil conditions, and management practices. Detrended yield anomaly is the appropriate target for a model whose inputs are also primarily year-specific factors.

The one-year lagged yield anomaly is included as an agronomic feature because it captures persistence in local production conditions and the carryover effects of management decisions or soil conditions from the prior growing season. Seed inventory,

soil compaction, residue conditions, and pest pressure can all carry forward in ways that influence the following year’s outcome, and the lag provides a data-driven representation of this dynamic without requiring explicit records of those intermediate mechanisms.

A binary crop indicator (corn = 1, soybean = 0) is included to allow the model to learn crop-specific responses to the same climate and soil conditions. Corn and soybean have distinct yield formation periods, drought sensitivity windows, and heat tolerance thresholds, and a single indicator allows those differences to be captured jointly without fitting separate models. Harvested acreage is retained as a normalizing variable for the conservation investment analysis in Chapter 6, where it converts practice funding to a per-acre basis comparable across counties of different sizes.

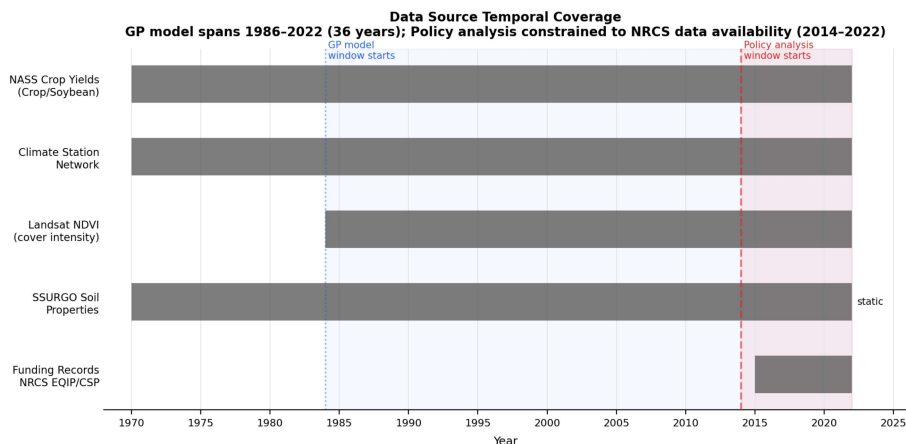


Figure 3.2: Temporal coverage of the five data sources used in the thesis, showing the overlap between climate, yield, NDVI, soil, and NRCS funding records.

3.3.1 USDA Census of Agriculture Data

County agricultural system classification is derived from USDA NASS Census of Agriculture county profile reports for 2022 [26]. Each county profile includes a “Share of Sales by Type” breakdown and livestock inventory estimates that together characterize whether a county’s agricultural economy is dominated by row-crop production, livestock production, or a mixed system. This information is used only as a scope filter for the yield-based policy analysis (see Section 6.3) and is not included as a feature in the Gaussian Process model.

3.4 Remote Sensing: Off-Season NDVI

Cover crop intensity is proxied using off-season Normalized Difference Vegetation Index (NDVI) extracted from Google Earth Engine. The off-season window is defined as November through March, when primary commodity crops are absent from the field and any vegetative signal reflects winter cover crops, cover crop residue, or early spring establishment. This approach follows the satellite-based cover crop detection methodology developed by Sun et al., who showed that multi-source imagery can track adoption trends at regional scale with sufficient accuracy to detect policy-driven increases in adoption rates.

Landsat 5/7/8 and MODIS Terra imagery are combined to construct annual county-level NDVI summaries. Cloud masking is applied using the quality assurance

(QA) bands associated with each product to exclude pixels with likely cloud contamination. County administrative polygons are used to spatially aggregate pixel-level observations to county means. The off-season NDVI value assigned to a given crop year is the mean of monthly mean NDVI values across the November-to-March window preceding that growing season.

This continuous proxy for vegetative cover is preferable to a binary enrollment indicator for several reasons. Conservation practice adoption is not uniform across a county’s fields; the proportion of enrolled versus non-enrolled acreage, the seeding density, and the temporal persistence of cover crop establishment all vary within a county from year to year. A mean NDVI signal captures that variation in a way that a binary variable cannot. It also captures non-enrolled land management practices such as winter wheat or other cover grasses that contribute to the vegetative signal without being formally tied to NRCS contracts.

3.5 Soil Data

Soil properties are extracted from the USDA SSURGO database via the Soil Data Access (SDA) web interface [13]. The query fetches component-level characteristics for each target county and then aggregates these to the county scale using component percentage as the weight. This ensures the relative importance of different soil types within a county is maintained and not all map units are treated equally.

The soil feature group includes available water capacity (AWC, volume fraction), organic matter content (weight percent), drainage class (encoded as continuous

score), a composite soil vulnerability index derived from drainage class and available water capacity, and county centroid latitude and longitude. These six variables form the soil and spatial feature group for the Gaussian Process model.

AWC and drainage class are critical soil properties in the Maumee Basin for different reasons. In dry years, available water capacity buffers crop water demand and delays the onset of stress; counties with high available water capacity are relatively insulated from short-duration drought stress. In wet years, drainage class dictates the rate at which water moves through the soil profile; counties that are poorly drained and rely on surface drainage may be subject to prolonged saturation of the root zone and subsequent reductions in yields due to low oxygen availability. The composite soil vulnerability index combines these two critical soil properties into a single feature that captures the county’s aggregate susceptibility to soil-driven yield stress.

Soil properties are considered temporally constant during the simulation period because the structural properties derived from SSURGO data, such as AWC, texture class, organic matter, and drainage, evolve on a geologic time scale, which is considered slow compared with the annual simulation window. County centroid coordinates are included as spatial features to allow the model to capture residual geographic structure in yield outcomes not fully explained by the tabulated soil attributes.

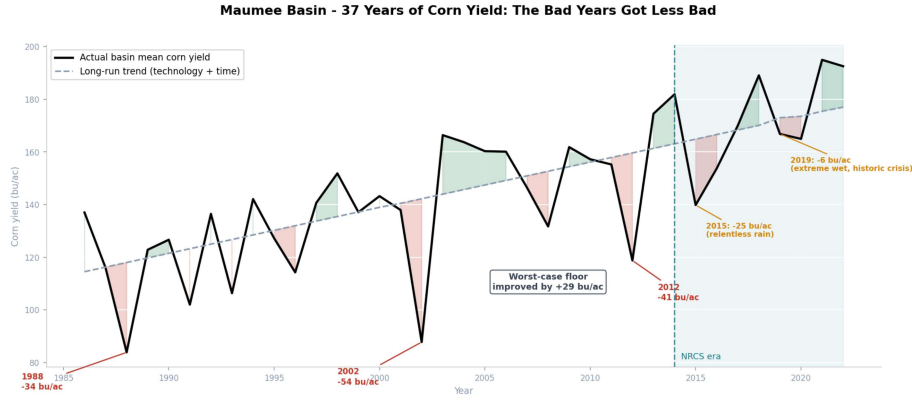


Figure 3.3: County-level yield anomaly trends for corn across the Maumee Basin study period.

3.6 NRCS Conservation Practice Funding Data

County-level NRCS practice funding records are obtained from USDA public program data for fiscal years 2014-2022 [16, 25]. The dataset spans six NRCS workinglands program streams: EQIP Farm Bill, EQIP IRA, RCPP-EQIP, CSP Farm Bill, CSP IRA, and the CSP Grassland Conservation Initiative (CSP-GCI). The dataset includes obligated dollars and practice instance counts aggregated by county, fiscal year, practice code, and conservation program. Practice names are parsed to extract the NRCS numeric code suffix, which allows records to be mapped to specific conservation practices: cover crop (340), nutrient management (590), no-till (329), residue management (344/345), conservation cover (327), and related yield-relevant soil health practices.

The resulting filtered funding data, normalized to per-acre terms using county harvested acreage, form the input to the Conservation Effectiveness Score developed in Chapter 6.

3.7 Feature Engineering

The final county-year feature matrix is constructed by merging all five data sources on a county-year-crop index. The resulting feature set is organized into three groups that correspond directly to the three GP sub-kernels: a six-dimensional climate vector, a three-dimensional agronomic vector, and a six-dimensional soil and spatial vector. This grouping is deliberate and reflects the distinct physical and temporal characteristics of the information in each group.

Several preprocessing transformations are applied before modeling. All continuous features within each domain group are standardized to zero mean and unit variance using training-set statistics only, preventing information leakage from test observations into the normalization procedure. Yield is detrended as described above. NRCS funding is converted to per-acre terms. Off-season NDVI is aggregated across the five-month window to reduce sensor noise and cloud contamination. County centroid coordinates are standardized to the same scale as other soil features so that they contribute to the GP covariance on a comparable basis.

Missing values in the yield record are handled by dropping county-year-crop observations with missing yield anomaly or missing lagged yield anomaly. Counties with fewer than fifteen usable observations after this filter are excluded from the

model entirely because GP fitting is unreliable with very sparse records. After all filtering, the final modeling dataset contains 1,095 county-year-crop observations across sixteen counties and thirty-seven years.

3.8 Dataset Summary

Table 3.1 summarizes the main data sources, their temporal coverage, and the features derived from each.

Table 3.1: Summary of data sources, temporal coverage, and derived features.

Source	Resolution	Coverage	Derived Features
NOAA NCEI	Station / day	1970–2022	SPEI-3, SPEI-6, SPEI-12, GDD, EHD, growing-season precipitation
USDA NASS	County / year	1970–2022	Yield anomaly, lagged yield anomaly, crop indicator, acreage
Landsat / MODIS	County / month	1984–2022	Off-season NDVI mean (cover intensity)
USDA SSURGO / SDA	County / static	Latest survey	AWC, OM, drainage score, soil vulnerability, latitude, longitude
NRCS practice data	County / fiscal year	2014–2022	Yield-relevant practice funding, practice counts, conservation indices

Chapter 4: The Domain-Decomposed Multi-Kernel Gaussian Process

4.1 Overview

A Gaussian Process (GP) defines a prior distribution over functions $f : \mathcal{X} \rightarrow \mathbb{R}$ such that for any finite collection of inputs $\{x_1, \dots, x_n\}$, the corresponding function values are jointly Gaussian [20]. Given an observed dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ where $y_i = f(x_i) + \varepsilon_i$ and $\varepsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the posterior predictive distribution at a new input x^* is

$$p(y^* | x^*, \mathcal{D}) = \mathcal{N}(\hat{\mu}, \hat{\sigma}^2), \quad (4.1)$$

where $\hat{\mu}$ and $\hat{\sigma}^2$ are closed-form functions of the kernel matrix and the training targets [20]. The predictive mean $\hat{\mu}$ serves as the point estimate of yield anomaly, and the predictive standard deviation $\hat{\sigma}$ provides the uncertainty estimate that later supports the Agricultural Risk Index and related policy indices.

The central modeling choice in this thesis is to use a domain-decomposed additive kernel structure rather than a single isotropic kernel over all input features. The full feature vector is partitioned into three disjoint subspaces:

$$x_c \in \mathbb{R}^6 \quad (\text{climate features}) \quad (4.2)$$

$$x_a \in \mathbb{R}^3 \quad (\text{agronomic features}) \quad (4.3)$$

$$x_s \in \mathbb{R}^6 \quad (\text{soil and spatial features}) \quad (4.4)$$

The total kernel is defined as the sum of three sub-kernels operating on these disjoint subspaces, together with a white-noise term representing measurement error and unexplained variance:

$$k(x, x') = k_c(x_c, x'_c) + k_a(x_a, x'_a) + k_s(x_s, x'_s) + k_{\text{noise}}. \quad (4.5)$$

This additive structure reflects the modeling assumption that climate stress, agronomic carry-over effects, and soil buffering capacity contribute in structurally distinct but complementary ways to county-level yield outcomes. It also preserves interpretability: each domain can be analyzed and selected separately rather than being absorbed into a single undifferentiated covariance. The design follows the additive Gaussian Process framework of [4].

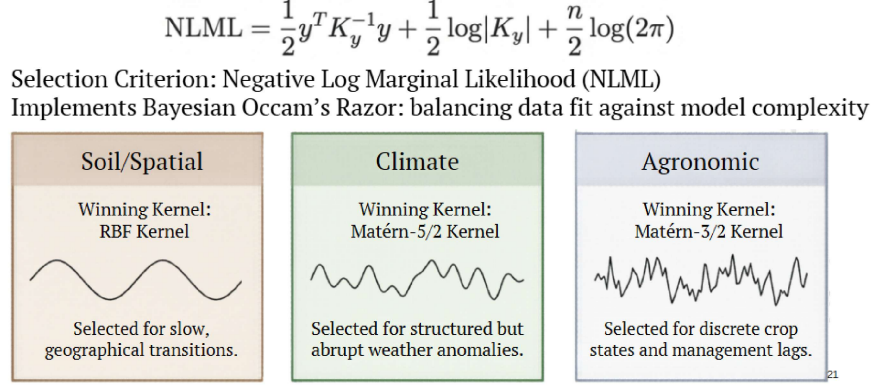


Figure 4.1: Conceptual structure of the domain-decomposed multi-kernel Gaussian Process. Separate kernels are assigned to climate, agronomic, and soil/spatial feature groups. The resulting covariance components are added to form the final GP prior. The design reflects the fact that each feature domain has a different smoothness structure and a different agronomic interpretation.

4.2 Kernel Selection by Negative Log Marginal Likelihood

Each sub-kernel family is selected from a candidate set by minimizing the Negative Log Marginal Likelihood (NLML) on the training data. The NLML is defined as

$$\mathcal{L}(\theta) = -\log p(y \mid X, \theta) = \frac{1}{2} y^\top K_y^{-1} y + \frac{1}{2} \log |K_y| + \frac{n}{2} \log 2\pi, \quad (4.6)$$

where $K_y = K + \sigma_n^2 I$ and θ denotes the full set of kernel hyperparameters [20]. This objective provides a principled Bayesian model selection criterion. The data-fit term $\frac{1}{2} y^\top K_y^{-1} y$ rewards kernels that explain the training targets well, while the complexity penalty term $\frac{1}{2} \log |K_y|$ penalizes kernels that can explain the data only

through highly concentrated probability mass over a restricted class of functions. This balance implements the Bayesian Occam’s Razor principle: among kernels that explain the data equally well, the simpler one is preferred.

Climate feature group. The candidate kernels are Matérn-3/2, Matérn-5/2, and RBF. Matérn-5/2 achieves the best NLML and is selected. This is appropriate because climate variables such as SPEI are structured but not infinitely smooth: they vary year to year in a way that is smoother than abrupt shocks but rougher than soil properties.

Agronomic feature group. The candidates are a Matérn-5/2, Matérn-3/2, and RBF. Matérn-3/2 achieves the best NLML and is selected. This is appropriate because the agronomic feature group contains variables with sharper year-to-year changes than soil properties: crop type is discrete, lagged yield anomaly can shift abruptly after shocks, and cover-intensity signals can vary with management decisions, making the group rougher than the slowly varying soil/spatial feature group.

Soil and spatial feature group. The candidates are RBF, Matérn-5/2, and Matérn-3/2. RBF achieves the best NLML and is selected. This result is consistent with the expectation that soil properties and county-level spatial coordinates vary smoothly across the landscape: soil texture, organic matter, and drainage class change gradually across neighboring counties in a way that an infinitely smooth

RBF can represent well. Unlike climate signals, soil properties do not exhibit abrupt annual transitions.

Table 4.1 reports the full NLML and test-set performance for all candidate kernels.

Table 4.1: Sub-kernel structure selection results. NLML is the primary model-selection criterion; RMSE and CRPS are reported on the held-out test set for reference. Lower NLML, RMSE, and CRPS are better. Bold indicates the selected kernel for each group.

Group	Kernel	NLML	RMSE	CRPS
Climate	Matérn-3/2	1325.66	11.39	6.47
	Matérn-5/2	1325.12	11.54	6.55
	RBF	1325.49	11.69	6.63
Agro	Matérn-5/2	1426.34	12.93	7.33
	Matérn-3/2	1362.76	11.89	6.60
	RBF	1366.53	12.20	6.74
Soil	RBF	1431.00	12.49	7.12
	Matérn-5/2	1433.82	12.49	7.12
	Matérn-3/2	1437.12	12.48	7.12

4.3 Data-Driven Hyperparameter Initialization

Standard GP implementations often initialize hyperparameters at arbitrary default values. That is acceptable for small examples but problematic for a county-level agricultural dataset where the optimization surface is non-convex and feature groups operate on different scales. This thesis therefore uses data-driven initialization throughout.

For each feature group, the initial length scale is set to the median pairwise Euclidean distance between training observations in the corresponding standardized feature subspace. This *median heuristic* places the model in a stable initialization regime in which approximately half of all training pairs have non-trivial covariance under the kernel - neither near zero (the kernel does not communicate between these points) nor near one (the kernel treats them as identical). This property has been studied formally and shown empirically to reduce sensitivity to initialization compared to uniform or random starting points [2, 17].

For the current dataset, the median pairwise distances in the standardized feature spaces are 2.64 for the climate group, 2.20 for the agronomic group, and 3.26 for the soil and spatial group. These values reflect the different dimensionality and spread of each feature group after standardization.

The signal variance is initialized at the empirical variance of the training target: $\sigma_f^2 = \text{Var}(y) = 246.67$. The noise variance is initialized at 10% of the signal variance: $\sigma_n^2 = 24.67$. This initialization encodes the reasonable prior that measurement error constitutes a minority of the total variance in county-level yield anomaly, which is dominated by structured climate and soil signals rather than noise.

A sensitivity analysis over a range of noise initializations from $0.01\times$ to $10\times$ the base estimate confirms that the final optimized solution is stable once the initialization is within the correct order of magnitude. Additional optimizer restarts beyond three produce negligible improvement in NLML, which indicates that the

data-driven initialization already places the optimizer near a well-conditioned local optimum. Seven restarts are used in the final model to ensure robustness against residual sensitivity.

Table 4.2 summarizes the initialization choices and their justifications.

Table 4.2: Hyperparameter initialization and sensitivity analysis. The final solution is robust to moderate perturbations of the noise initialization and optimizer restart count.

Quantity	Value	Interpretation	Used in final model
Climate length scale	2.64	Median pairwise distance	Yes
Agronomic length scale	2.20	Median pairwise distance	Yes
Soil length scale	3.26	Median pairwise distance	Yes
Signal variance	246.67	Empirical target variance	Yes
Noise variance	24.67	10% of signal variance	Yes
Optimizer restarts	7	Robust against local optima	Yes

4.4 Final Learned Kernel

After optimization by L-BFGS-B over the log-hyperparameter space starting from the data-driven initialization, the final learned kernel is

$$k = 2.27^2 \cdot k_{\text{Mat5/2}}(\ell = 10) + 1.11^2 \cdot k_{\text{Mat3/2}}(\ell = 10) + 1.87^2 \cdot k_{\text{RBF}}(\ell = 5.31) + k_{\text{White}}(\sigma_n^2 = 0.558). \quad (4.7)$$

The climate sub-kernel has the largest variance, approximately 5.15, indicating that climate contributes the strongest signal to county-level yield variation. This

is consistent with the well-established finding that growing-season drought and heat stress explain a large share of interannual yield variation in Midwestern corn-soybean systems. The soil sub-kernel follows, with variance approximately 3.50, reflecting substantial spatial structure in yield productivity associated with soil water capacity and drainage. The agronomic sub-kernel is smaller, with variance approximately 1.23, capturing the marginal contribution of cover intensity, crop type, and prior-year conditions beyond what climate and soil account for.

The learned noise term is small ($\sigma_n^2 = 0.558$) relative to the summed signal variance of approximately 9.98. This high signal-to-noise ratio is an important result because the policy indices in Chapter 6 depend on the predictive distribution being informative enough to distinguish between counties on the basis of their risk level. A model in which noise dominated the signal would produce wide, nearly identical prediction intervals that could not support meaningful county-level ranking.

4.5 Baseline Models and Experimental Setup

The multi-kernel GP is compared against deterministic baselines such as Linear Regression, Ridge Regression ($\alpha = 1.0$), Random Forest (200 trees), Gradient Boosting (200 estimators), and Support Vector Regression. The probabilistic baseline is a single-kernel GP variants using an RBF kernel, a Matérn-3/2 kernel, or a Matérn-5/2 kernel, depending on the variant in Table 5.1. All models are trained on data from 1986-2017 and tested on 2018-2022 (a five-year prospective holdout).

The deterministic baselines are evaluated using RMSE and R^2 . The GP models are evaluated using RMSE, Continuous Ranked Probability Score (CRPS), and the empirical coverage of the 90% prediction interval (Coverage@90). This asymmetric reporting is intentional: the thesis is primarily interested in probabilistic performance, and the deterministic baselines are included as reference points rather than as directly competing methods.

For LOCO, the comparison is restricted to GP variants because the ARI policy application requires predictive distributions rather than point forecasts.

4.6 Model Evaluation Strategy

Two forms of validation are used in Chapter 5. First, Leave-One-County-Out (LOCO) cross-validation tests whether the model generalizes to counties excluded from training, simulating the deployment scenario of applying the model to a new county not seen during fitting. Second, a prospective temporal horizon analysis tests whether the model retains useful predictive skill for observations beyond the end of the training window, which is directly relevant to its use as a planning tool for future growing seasons.

This evaluation design is intentionally strict. A county-level model can appear accurate in-sample by memorizing county-specific structure that does not generalize. LOCO directly tests generalizability. Similarly, a model that fits historical data well may lose calibration under future conditions when climate patterns shift or when

the model encounters years outside its training distribution. The temporal horizon analysis tests that property explicitly.

4.7 Summary

The main technical contribution of this chapter is a county-level multi-kernel Gaussian Process that separates climate, agronomic, and soil/spatial signal structures using sub-kernels selected by Negative Log Marginal Likelihood, with hyperparameters initialized from data-driven heuristics. The resulting model is both predictive and interpretable, and the final learned kernel structure provides interpretable quantification of how much each feature domain contributes to the county-level yield anomaly signal. This probabilistic foundation supports the policy analysis developed in Chapter 6.

Chapter 5: Model Evaluation

5.1 Baseline Comparison

Table 5.1 reports the baseline comparison under the temporal split, with training on 1986-2017 and testing on the five-year holdout period 2018-2022.

Table 5.1: Baseline comparison under temporal split (test: 2018-2022, $n = 111$ observations). Lower RMSE and CRPS are better; higher R^2 and Coverage@90 are better. R^2 is not reported for GP models because the primary evaluation criteria are probabilistic (CRPS and Coverage@90)

Model	RMSE	R^2	CRPS	Cov@90
Linear Regression	13.61	-0.305	—	—
Ridge Regression	13.61	-0.305	—	—
Random Forest	12.19	-0.047	—	—
Gradient Boosting	13.60	-0.303	—	—
SVR	11.61	+0.050	—	—
GP-RBF	12.80	—	7.000	0.856
GP-Matern32	12.55	—	6.850	0.847
GP-Matern52	12.66	—	6.913	0.856
Multi-Kernel GP (this work)	12.54	—	6.850	0.856

Table 5.1 shows that the temporal holdout period is genuinely difficult: most models achieve negative or near-zero R^2 , indicating that naive extrapolation of historical structure is insufficient for 2018-2022. SVR obtains the lowest point-prediction RMSE (11.61 bu/ac), but it does not provide a predictive distribution, which limits its usefulness for policy deployment. Among the GP models, the multi-kernel GP attains the lowest temporal-split RMSE (12.54), narrowly ahead of GP-Matern32 (12.55), while CRPS is tied at 6.850 between the two models.

The main advantage of the multi-kernel GP is not best temporal-split RMSE, but interpretability and calibrated policy-oriented uncertainty. A single-kernel GP such as GP-RBF is the natural ablation of the multi-kernel design because it removes the domain decomposition and collapses the three feature groups into one covariance structure. In contrast, the multi-kernel GP assigns separate covariance functions to climate, agronomic, and soil feature groups. This matters because the relevant covariance structure differs across domains: climate effects accumulate over different time scales than soil properties or agronomic management.

This domain-aware structure is also supported by the LOCO results, which are the most relevant for deployment to new counties. On LOCO, the multi-kernel GP achieves $\text{RMSE} = 12.56$ bu/ac, $\text{CRPS} = 6.95$ bu/ac, and $\text{Coverage@90} = 91.9\%$, improving on the single-kernel GP-RBF baseline on CRPS and providing calibrated uncertainty for policy use.

5.2 Leave-One-County-Out Evaluation

The primary validation protocol is Leave-One-County-Out (LOCO) cross-validation. In each fold, the model is trained on all counties except one and evaluated on the withheld county. This evaluation is the most stringent test of county-level generalizability because it directly simulates deployment to a county that was not present during training.

Table 5.2 reports per-county LOCO results for the multi-kernel GP.

Table 5.2: Leave-One-County-Out evaluation results for the multi-kernel GP. Coverage@90 is the empirical fraction of test observations falling inside the model’s 90% prediction interval. The final row reports the basin-level mean.

County	n	RMSE	CRPS	Cov@90
Allen	73	18.72	10.33	0.877
Auglaize	74	13.09	7.18	0.892
Defiance	73	10.62	5.96	0.945
Fulton	73	8.72	5.30	0.986
Hancock	71	12.21	6.80	0.901
Hardin	44	16.38	8.93	0.955
Henry	60	11.99	6.38	0.933
Lucas	72	9.58	5.52	0.944
Mercer	74	14.86	7.84	0.905
Ottawa	60	13.94	7.85	0.867
Paulding	73	11.88	6.45	0.904
Putnam	60	13.87	7.38	0.883
Sandusky	74	10.42	6.08	0.973
Shelby	74	13.56	7.31	0.892
Williams	60	10.33	5.92	0.950
Wood	72	10.76	5.95	0.903
Mean	67.9	12.56	6.95	0.919

The basin-level mean Coverage@90 of 0.919 indicates slight over-coverage relative to the nominal 90% target. Conservative calibration (where actual coverage exceeds nominal coverage) is preferable to overconfident intervals in a policy-support context. An interval that is slightly wider than necessary means that the model is less likely to signal false confidence in a yield forecast, which matters when the output is used to prioritize conservation outreach.

Performance varies across counties in ways that have interpretable explanations. Fulton County achieves the strongest calibration and one of the lowest RMSE values, consistent with its highly homogeneous row-crop character and well-constrained soil and climate conditions. Allen County shows the highest RMSE, consistent with its more diverse agricultural structure and transitional position at the eastern edge of the basin where soil types and drainage patterns are more variable. Mercer County’s elevated RMSE is also consistent with its livestock-dominant character, which means that the yield anomaly signal in that county is less tightly linked to the climate and soil features that drive the model elsewhere.

5.3 Calibration and Probabilistic Performance

A probabilistic model is useful for policy applications only if its uncertainty is calibrated as well as sharp. For a nominal coverage level p , a well-calibrated model should contain observed values within its $p\%$ prediction interval approximately p fraction of the time across the entire test set. In this thesis, calibration is assessed

using the empirical Coverage@90 metric and the Continuous Ranked Probability Score (CRPS).

CRPS evaluates the entire predictive distribution simultaneously:

$$\text{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} \left[\hat{F}(z) - \mathbf{1}(z \geq y) \right]^2 dz, \quad (5.1)$$

where $\hat{F}$ is the predictive cumulative distribution function and y is the observed value. For a Gaussian predictive distribution with mean $\hat{\mu}$ and standard deviation $\hat{\sigma}$, CRPS has a closed-form expression that penalizes both poor calibration (incorrect coverage) and low sharpness (unnecessarily wide intervals). Lower CRPS indicates a better predictive distribution.

The basin-mean CRPS of 6.95 bu/ac under LOCO improves on the single-kernel GP-RBF baseline, confirming that the domain-decomposed kernel structure improves the quality of the predictive distribution, not only point accuracy. The consistency between empirical Coverage@90 (0.919) and the nominal target (0.90) is an important validation result, because it indicates that the model’s uncertainty estimates err on the side of caution rather than overconfidence, which is the appropriate direction for a decision-support tool.

5.4 Performance by Year Type

Table 5.3 reports LOCO performance stratified by year type.

The model achieves its best performance in dry years (RMSE = 11.39, CRPS = 6.30), which is an important result for the policy application. Dry years are among

Table 5.3: LOCO performance stratified by year type. Dry years: SPEI-12 < -0.5 ; wet years: SPEI-12 $> +0.5$; normal years otherwise.

Year Type	n	RMSE	CRPS	Cov@90
Dry	205	11.39	6.30	0.941
Normal	565	12.96	7.00	0.912
Wet	317	13.11	7.13	0.918

the most actionable conditions for conservation intervention because yield shortfalls are most likely to motivate outreach, and conservation targeting is most urgent. The model’s stronger performance in those years means that the ARI index is most reliable exactly when it is most needed.

The relatively weaker performance in wet years (RMSE = 13.11) is expected. Waterlogging and excess moisture effects are driven by within-county variation in tile drainage completeness, field-level topography, and subsurface hydrology that are not captured by the county-level feature set. No county-level climate index fully captures the differential within-county response to excessive soil moisture that depends on field-scale drainage infrastructure. This is a genuine and acknowledged limitation of the county-level modeling framework.

5.5 Temporal Horizon Evaluation

A prospective temporal horizon analysis evaluates how much the model’s predictive quality degrades as the number of years forecasted beyond the training window

increases. For a horizon of h years, the model is trained on data up to $2022 - h$ and evaluated on the subsequent h years.

Table 5.4 reports the results across horizons from $h = 1$ to $h = 5$ years.

Table 5.4: Prospective temporal horizon evaluation. Each row represents a forecast horizon h years, with training restricted to years up to $2022 - h$ and testing on the subsequent h years.

Horizon h	n	RMSE	CRPS	Cov@90
$h = 1$	24	16.34	9.59	0.708
$h = 2$	46	14.47	8.24	0.783
$h = 3$	69	12.32	6.82	0.870
$h = 4$	87	11.46	6.31	0.885
$h = 5$	111	12.75	6.97	0.856

The most important result from this analysis is the stability of model performance at the four-year horizon. At $h = 4$, the model achieves $\text{RMSE} = 11.46$ and $\text{CRPS} = 6.31$, both of which are comparable to the LOCO mean across all counties. This indicates that the GP retains useful predictive structure several years beyond the training window, supporting its use as a planning-oriented outlook tool rather than only a historical fit.

The one-year horizon performs considerably worse than the longer horizons ($\text{RMSE} = 16.34$). This counterintuitive pattern reflects the specific difficulty of the $h = 1$ test period rather than a structural failure of the model: the single year held out at $h = 1$ is 2022, which was an unusual growing season in the basin. The shorter

training window available at $h = 1$ also increases model uncertainty, which is reflected in the lower Coverage@90 of 0.708 (the model’s intervals are too narrow for this anomalous year). The decreasing-then-stabilizing pattern from $h = 3$ to $h = 5$ is the behavior expected of a properly calibrated model: performance degrades as historical context is removed, but does not collapse entirely.

These results support the claim made in the thesis statement: the model produces calibrated predictions that retain useful planning skill up to approximately four years beyond the training window, establishing a practical horizon for early-season agricultural outlook applications.

5.6 Summary

The four evaluation protocols consistently identify the domain-decomposed multi-kernel GP as the most appropriate model for policy deployment. The temporal-split results show that SVR achieves the lowest point-prediction RMSE, but it does not provide predictive uncertainty. On LOCO, the multi-kernel GP achieves RMSE = 12.56 bu/ac, CRPS = 6.95 bu/ac, and Coverage@90 = 91.9%, improving on the single-kernel GP-RBF baseline on CRPS. Its strong coverage calibration and robustness in dry years indicate reliability when conservation targeting is most urgent. Altogether, these findings satisfy RQ1 and RQ2 and establish the multi-kernel GP as a reliable technical foundation for the policy analysis in the following chapter.

Chapter 6: Conservation Policy Analysis

6.1 From Model Outputs to Policy Metrics

This chapter translates the Gaussian Process output and NRCS practice funding records into a county-level conservation targeting framework comprising two constructed indices. Because no existing metric captures yield risk, accumulated conservation investment, and resilience jointly at county scale, the metrics are developed explicitly for this thesis. They are intended as interpretable decision-support tools for NRCS district conservationists and Technical Service Providers, not as universal indicators.

The metrics serve distinct and complementary roles. The *Agricultural Risk Index* (ARI) summarizes predicted vulnerability by combining expected yield shortfall with predictive uncertainty into a county-year risk score. The *Conservation Effectiveness Score* (CES) summarizes accumulated conservation investment over a five-year horizon using lagged, acreage-normalized funding with temporally decaying weights.

Together, these metrics allow the analysis to distinguish between counties that are highly vulnerable and under-supported, counties that are vulnerable but already receiving sustained conservation attention, and counties where accumulating investment may be approaching the threshold for measurable benefit. That typology is the core policy output of this thesis.

6.2 Agricultural Risk Index

The Agricultural Risk Index combines two outputs of the GP predictive distribution: the predicted yield shortfall and the predictive uncertainty. The goal is to create a transparent, interpretable county-year vulnerability score that reflects both expected downside and confidence in that expectation.

$$\text{ARI}_{i,t} = \frac{1}{2}\tilde{r}_{i,t}^{\mu} + \frac{1}{2}\tilde{r}_{i,t}^{\sigma}, \quad (6.1)$$

where $\tilde{r}_{i,t}^{\mu}$ is the min-max normalized value of $-\hat{\mu}_{i,t}$ (so that larger predicted shortfalls produce higher ARI values) and $\tilde{r}_{i,t}^{\sigma}$ is the min-max normalized value of $\hat{\sigma}_{i,t}$ (so that greater predictive uncertainty also produces higher ARI values). Both components are normalized across all county-year-crop observations, and the resulting ARI is rescaled to lie in $[0, 1]$.

The equal 0.5–0.5 weighting of shortfall and uncertainty encodes a deliberate policy assumption. From a decision-making standpoint, a county with a large but certain predicted yield deficit and a county with a highly uncertain but potentially severe deficit are both policy-relevant. The first needs intervention because the risk

is real; the second needs intervention because the risk cannot be ruled out. Treating both dimensions equally means that ARI is not just a forecast of poor performance - it is a combined vulnerability measure. The ARI therefore captures counties that are clearly at risk and counties that are *ambiguously* at risk, both of which may warrant outreach.

ARI values near 0 indicate county-year conditions that the model predicts to be near or above average with relatively high confidence. Values near 1 indicate county-year conditions combining low predicted yield with high uncertainty. This interpretation is stable across the range of years and counties in the study.

6.3 County Agricultural System Classification

Before applying the yield-based policy framework, the thesis classifies all sixteen Maumee Basin counties by agricultural system type using USDA NASS 2022 County Profile reports. This step is necessary because a yield-risk framework built on corn and soybean anomalies is not methodologically appropriate for counties whose agricultural economy is dominated by livestock production. Applying ARI-based conservation targeting uniformly to all counties would direct outreach toward counties where yield anomaly is not a reliable proxy for agricultural vulnerability.

Classification is based on two dimensions of the county profile: the “Share of Sales by Type” breakdown, which reports the percentage of total agricultural sales

attributable to crops versus livestock and livestock products, and the livestock inventory totals, which together indicate whether a county’s predominant agricultural activity is row-crop production or animal production.

Counties are classified as row-crop-dominant if crop sales constitute the clear majority of total agricultural sales and the livestock inventory is low relative to harvested acreage. Counties are classified as Livestock-dominant if livestock and livestock product sales dominate and the animal unit equivalent per harvested acre is high. Counties that fall near the boundary between these two categories are classified as mixed.

Table 6.1 reports the classification for all sixteen counties.

For the policy analysis, the county set is restricted to row-crop-dominant counties with sufficient NRCS coverage in the 2014-2022 funding data. Mixed and Livestock-dominant counties are included with the caveat that their ARI reflects row-crop yield risk specifically, not total agricultural vulnerability.

6.4 Conservation Effectiveness Score

The Conservation Effectiveness Score is a constructed measure of accumulated conservation investment intended to approximate the persistence of soil health and management benefits over time. It weights recent NRCS funding more heavily than older funding and normalizes by harvested acreage to produce a per-acre investment measure that is comparable across counties of different agricultural scales.

Table 6.1: County agricultural system classification based on USDA NASS 2022 County Profile reports. Classification reflects share of sales by type and livestock inventory indicators from the official county profiles.

County	Category	Crop %	Livestock %
Allen	Mixed	58	42
Williams	Row-crop-dominant	72	28
Defiance	Mixed	56	44
Paulding	Mixed	47	53
Mercer	Livestock-dominant	20	80
Fulton	Row-crop-dominant	77	23
Henry	Row-crop-dominant	87	13
Putnam	Row-crop-dominant	63	37
Auglaize	Mixed	57	43
Shelby	Mixed	52	48
Lucas	Row-crop-dominant	94	6
Hancock	Row-crop-dominant	82	18
Hardin	Mixed	51	49
Ottawa	Row-crop-dominant	92	8
Sandusky	Row-crop-dominant	92	8
Wood	Row-crop-dominant	72	28

$$\text{CES}_{i,t} = \sum_{j=1}^5 w_j \cdot \frac{\text{PAI}_{i,t-j}}{\text{acres}_{i,t-j}}, \quad (6.2)$$

where $\text{PAI}_{i,t-j}$ is the total yield-relevant NRCS practice investment in county i during year $t - j$, $\text{acres}_{i,t-j}$ is the county’s mean harvested acreage in that year, and the lag weights are $w_j = \{0.35, 0.25, 0.20, 0.12, 0.08\}$ from $j = 1$ to $j = 5$.

The decaying weight structure reflects the empirical observation that conservation benefits emerge gradually rather than immediately and that older investments contribute less to current soil condition than more recent ones [15, 14, 27]. The most

recent year receives the highest weight (0.35) because recently implemented practices are the most directly connected to current soil health conditions. Investments from five years ago receive a weight of 0.08, acknowledging that while multi-season persistence is real, the marginal contribution of older practices to current resilience diminishes as practices may have been disadopted, modified, or superseded. The five-year horizon matches CEAP evidence that soil-health benefits typically emerge over a 3-5 year window [27].

A sensitivity check over CES lag windows of three and seven years and over uniform versus front-loaded weight schemes confirms that 14 of 15 in-scope counties retain the same CAG quadrant assignment across all combinations tested. The one exception is a border county that shifts between Resilient-Low-Need and Resilient-Supported depending on the window length, but does not enter either high-ARI quadrant under any scheme.

Figure 6.1 illustrates the lag weight structure.

CES is min-max normalized to $[0, 1]$ across all in-scope county-year observations so that it can be used directly in the Conservation-Agriculture Gap classification on a common scale with ARI.

6.5 Policy Event Study

The event study examines county-level yield anomaly outcomes relative to the first year in which a county receives yield-relevant NRCS practice funding. This within-county design avoids the confounding that would arise from comparing funded

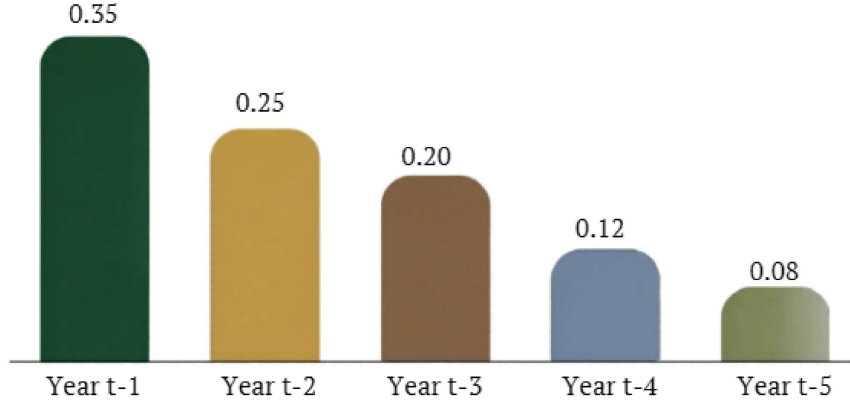


Figure 6.1: Lag weights used in the Conservation Effectiveness Score. The scheme assigns the greatest credit to the most recent year and progressively discounts older investments, reflecting the delayed accumulation of soil health and conservation practice benefits.

and unfunded counties directly, since those groups differ along many dimensions. Instead, each county serves as its own control, and the event study traces the county's deviation from its own pre-funding trend around the funding onset.

Let $T_{i,t} = t - t_i^*$, where t_i^* is the first year of yield-relevant NRCS funding for county i . For each event time $T \in \{-2, -1, 0, +1, \dots, +6\}$, the mean yield anomaly is computed across all in-scope counties at that relative time, with bootstrap confidence intervals (1,000 replications) used to quantify uncertainty.

The event-study results show a clear pattern of delayed benefit emergence. The most important finding is that yield anomaly shows its strongest positive deviation from the pre-event baseline approximately four years after initial funding, with a mean estimated recovery of +12.3 bu/ac relative to the pre-funding mean (95% CI:

8.2–16.9 bu/ac). This lag is consistent with the biological and agronomic expectation that soil health improvements from cover crops, reduced tillage, and nutrient management require multiple growing seasons of consistent practice application before becoming visible in the yield record [15, 14, 27].

The early event period also shows evidence of the selection effect that motivates conservation enrollment. In the two years prior to first funding ($T = -2$ and $T = -1$), county yield anomalies are slightly below their historical mean, consistent with counties entering the program after experiencing stress. This pattern is expected in any voluntary program where producer motivation to enroll is partly driven by recent negative experience. It should not be interpreted as a consequence of program enrollment but as its context. The absence of an immediate improvement at $T = 0$ also aligns with the designed lag structure of the CES, which discounts the most recent year less than older years but still does not assume immediate effects.

6.6 Conservation-Agriculture Gap Quadrant

The Conservation-Agriculture Gap (CAG) quadrant classifies in-scope row-crop-dominant counties by their position relative to the basin median values of ARI and CES. Counties are placed into four categories that carry distinct policy interpretations.

Priority target (high ARI, low CES). These counties show elevated yield risk but have below-median accumulated conservation investment. They represent the

clearest candidates for additional NRCS outreach, increased contract allocation, and focused technical assistance. Conservation capital has not yet been built up here to a level that could moderate predicted shortfalls.

Supported but struggling (high ARI, high CES). These counties have received above-median conservation investment but still show elevated ARI scores. Two interpretations are possible: either the conservation practices are implemented but not yet mature enough to have reduced risk visibly, or the current practice mix is not optimal for the specific stress drivers affecting those counties. Either way, these counties may benefit from a practice mix review and continued support rather than purely intensified investment.

Resilient and supported (low ARI, high CES). These counties show both relatively low risk and strong conservation history. They represent the intended outcome of sustained conservation investment: lower vulnerability paired with accumulated soil health capital. These counties can serve as regional benchmarks for what effective conservation implementation looks like over time.

Lower need (low ARI, low CES). These counties do not currently appear to require urgent conservation intervention on the basis of risk-adjusted yield vulnerability. They may be naturally resilient due to favorable soil conditions or consistently moderate climate, and they do not appear to be under-served by NRCS in a way

that is currently creating measurable risk. They remain useful as comparison cases for regional planning.

Figure 6.2 illustrates the quadrant classification.

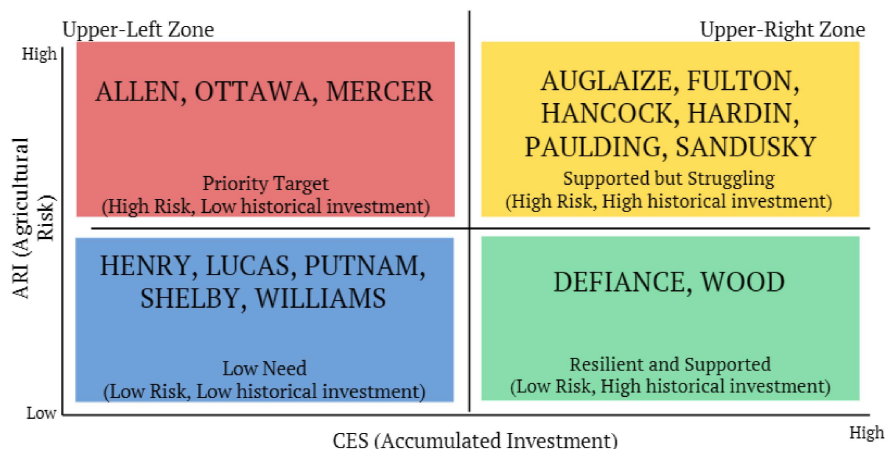


Figure 6.2: Conservation-Agriculture Gap quadrant classification. Counties are positioned by their normalized ARI and CES values relative to basin medians, distinguishing priority counties, supported-but-struggling counties, resilient counties, and lower-need counties.

6.7 Interpretation of County Classes

Among the in-scope row-crop-dominant counties, the quadrant structure identifies distinct policy responses not visible from funding totals alone. Priority counties combine high predicted vulnerability with limited conservation accumulation, making them the clearest candidates for immediate outreach, stronger technical assistance, and expanded financial support.

Supported-but-struggling counties present a more nuanced challenge. These counties have received investment, but risk remains elevated, suggesting a need for continued support and review of practice implementation quality. The event study’s T+4 recovery pattern is directly relevant here because counties with recent investment may still be approaching the period of maximum return.

Resilient-plus-supported counties illustrate the intended outcome of conservation programming: lower risk paired with sustained investment. These counties can serve as regional benchmarks and outreach examples for how multi-year conservation support may coincide with reduced vulnerability over time.

6.8 Summary

This chapter translated probabilistic GP outputs and NRCS funding records into two interpretable policy metrics and a county classification framework. ARI measures county-year vulnerability from the predictive distribution. CES summarizes the temporal accumulation of conservation investment.

The event study provides evidence consistent with a T+4 recovery pattern consistent with the lag structure built into CES, validating the choice of a five-year weighting window. The CAG quadrant converts continuous ARI and CES scores into an actionable county typology for NRCS planning. Together, these results provide evidence that the integrated framework produces policy-relevant signals that are grounded in the predictive model rather than in arbitrary thresholds, and that

those signals are consistent with the agronomic and policy literature on conservation benefit dynamics.

Chapter 7: Conclusion

7.1 Summary of Contributions

This thesis developed and validated an integrated framework for probabilistic agricultural risk assessment and conservation policy evaluation in the Maumee River Basin. The work addressed a gap that sits at the intersection of machine learning, agricultural science, and conservation policy: the absence of a county-level, uncertainty-aware analytical tool that can connect GP-derived yield risk to NRCS investment history in an interpretable and scope-bounded way.

First contribution: The domain-decomposed multi-kernel GP. The model assigns separate covariance functions to three disjoint feature domains (climate, agronomic, and soil) with sub-kernel structure selected by Negative Log Marginal Likelihood and hyperparameters initialized from data-driven heuristics. Under Leave-One-County-Out evaluation, it achieves $\text{RMSE} = 12.56 \text{ bu/ac}$, $\text{CRPS} = 6.95 \text{ bu/ac}$, and $\text{Coverage@90} = 91.9\%$. Among GP variants evaluated on the temporal split, the multi-kernel GP provides competitive calibration and interpretable domain-separated

uncertainty estimates. At the four-year prospective horizon, RMSE and CRPS remain comparable to the LOCO mean, establishing a practical planning window.

Second contribution: The policy evaluation framework. Two composite indices (ARI, CES) translate the GP’s predictive distribution and NRCS funding records into county-level policy signals. ARI quantifies combined vulnerability from expected shortfall and predictive uncertainty. CES summarizes the five-year accumulation of yield-relevant conservation investment using temporally decaying weights calibrated to the empirically documented benefit onset timeline. An event study provides observational evidence consistent with a T+4 yield recovery of approximately +12.3 bu/ac (95% CI: 8.2–16.9 bu/ac), providing county-level evidence consistent with the delayed conservation benefit emergence established in the field-scale literature.

Third contribution: The county agricultural system classifier. USDA Census of Agriculture county profile data are used to classify the sixteen Maumee Basin counties by agricultural system type, distinguishing row-crop-dominant counties from livestock-dominant and mixed counties. This classifier ensures that the yield-based risk framework is applied only to counties where corn and soybean yield anomaly is a meaningful proxy for agricultural vulnerability. The classifier is based entirely on publicly available data and is reusable for other watersheds where similar scope questions arise.

Together, these three contributions form a calibrated, scope-bounded, investment-aware framework for county-level agricultural risk assessment and conservation policy evaluation that is designed to be both scientifically defensible and practically usable.

7.2 Limitations

Several limitations of this work should be acknowledged clearly.

County-level aggregation. The GP model operates on county-level aggregates, which necessarily mask within-county heterogeneity in soil properties, management practices, and microclimate. In a large, internally diverse county, the county mean yield anomaly may not reflect the experience of any individual farm. Field-level predictions would require field-scale management records (planting dates, variety selection, nutrient application histories) that are not publicly available at the spatial and temporal extent of this study.

CES lag weight calibration. The specific decay weights $\{0.35, 0.25, 0.20, 0.12, 0.08\}$ in the CES are calibrated to the literature-documented accumulation timeline but are not estimated directly from the county-level data in this study. The five-year window and decaying structure are justified by the CEAP evidence, but the specific numerical weights represent an operationalization of domain knowledge rather than

a learned parameter. Sensitivity analysis shows that the CAG quadrant classification is stable across plausible weight perturbations, but the weights remain a design choice.

Livestock-dominant scope boundary. The livestock-dominant county excluded from the primary policy analysis represents a distinct conservation challenge, particularly around manure nutrient management, animal waste infrastructure, and phosphorus export pathways that differ fundamentally from row-crop nutrient losses. A livestock-adapted extension of the policy framework would be needed to represent those counties appropriately, and that extension is beyond the scope of this thesis.

Temporal extent of NRCS funding data. Practice funding data are available only from 2014, limiting the policy analysis to approximately nine years. This constrains the event study’s ability to examine long-term benefit trajectories beyond $T + 6$ and reduces the sample of counties whose first funding year falls early enough to generate a full event-time window. A longer funding record and a more rigorous quasi-experimental design would strengthen the causal interpretation of the event study.

Computational scalability. Exact GP inference scales cubically with training set size ($\mathcal{O}(n^3)$). The current dataset of approximately 1,095 county-year observations is manageable on standard hardware, but extending the framework to larger basins, finer spatial resolution, or substantially longer time series would require sparse or

approximate GP methods. This is a known property of exact GP implementations and is a practical constraint for scaling rather than a conceptual limitation.

7.3 Future Work

Five directions are identified as natural extensions of this work.

Field-scale integration. Coupling the county-level GP with field-scale precision agriculture data (cover crop seeding records, soil health scorecard observations, or NRCS practice implementation reports) would enable practice-specific benefit attribution and substantially improve the granularity of conservation targeting. As producer record-keeping becomes more digital and NRCS field data collection systems expand, the data infrastructure for this extension is becoming more feasible.

Livestock-adapted policy framework. Developing a parallel CES and risk index specifically for livestock-dominant counties using manure nutrient loading estimates, herd size trends, and Comprehensive Nutrient Management Plan adoption data would extend the conservation policy framework to the full Maumee Basin. The livestock sector is a significant contributor to phosphorus export in parts of the basin, and those counties deserve a comparably rigorous analytical framework.

Phosphorus loading integration. The current framework uses yield risk as a proxy for agronomic conservation need. Directly modeling dissolved reactive phosphorus (DRP) export from Maumee watershed monitoring stations as an additional

outcome would connect the yield-risk signal to the water-quality outcomes that motivate basin-wide conservation policy. Integrating load monitoring data with the GP risk framework would allow a more direct test of whether counties with high ARI or low CES also show elevated DRP contributions.

Online model updating. The GP framework could be extended to an incremental learning setting where new NASS yield observations are assimilated annually after each growing season. With suitable sparse GP approximations or sequential posterior updating, the county-level predictive distribution could be refreshed without full retraining. This would allow the ARI indices to be updated each fall as new yield data become available, supporting annually refreshed conservation planning cycles.

Uncertainty communication for practitioners. An open and practically important research question is how to communicate calibrated uncertainty estimates effectively to non-technical NRCS conservationists and Technical Service Providers who are not trained in probabilistic reasoning. The current framework produces well-calibrated intervals, but whether those intervals change outreach decisions in practice depends on how they are presented. User testing of a policy dashboard with NRCS field staff would provide evidence on whether prioritization influences resource allocation decisions and how interval width should be communicated to practitioners.

7.4 Conclusion

The Maumee River Basin faces a structural challenge that many nutrient-impaired agricultural watersheds share: conservation funding exists to address agricultural phosphorus runoff and yield vulnerability, but the allocation of that funding is not systematically guided by where conservation need is greatest, where prior investment may already be generating benefits, or where a yield-based framework is and is not appropriate. This thesis develops a quantitative framework for addressing that challenge at the intersection of computational machine learning, agricultural data science, and conservation policy.

The domain-decomposed multi-kernel Gaussian Process provides calibrated uncertainty estimates that make the downstream risk indices defensible. The CES metric provides the temporal lens that transforms ARI from a forecast into an actionable prioritization signal. The agricultural system classifier prevents the framework from being misapplied to counties where its underlying assumptions do not hold.

Taken together, these contributions form a framework that is calibrated, scope-bounded, and investment-aware. It is not a replacement for the field expertise of district conservationists and Technical Service Providers, but it is designed to give those practitioners a principled quantitative signal that complements their judgment. The question of where conservation dollars are most needed has always been difficult to answer rigorously. This thesis is an attempt to make that answer more data-driven and more transparent.

The same design principles (domain-decomposed probabilistic modeling, lagged investment accumulation, explicit scope validation) can be adapted to other nutrient-impaired agricultural watersheds in the United States and elsewhere that face similar questions about how to allocate limited conservation resources to where they can produce the greatest benefit.

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